Tô Gia Bảo

Data Engineer

Thu Duc, Ho Chi Minh City, Vietnam

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Summary

Data Engineer building batch and streaming pipelines, lakehouse platforms, and AI-assisted analytics tools with Python and SQL. At Timo Digital Bank, delivered reliable pipelines with dbt, self-service data discovery tools, and observability solutions across Airflow, Amazon EMR, AWS Glue, Athena, and S3. Hands-on with Spark, PostgreSQL, MongoDB, Docker, and Terraform, with a focus on trustworthy, analytics-ready data for business-critical decisions.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Built batch pipelines syncing MariaDB into a landing–curated–analytics lakehouse with Amazon EMR, AWS Glue, PII masking, and dbt, delivering reliable analytics-ready data for stakeholders.
- Optimized a high-volume S3 table with Parquet ZSTD-3 compression and a 128 to 512 MiB file-size contract, saving 38.4 TB (45%) and reducing object count by 80% across 56.8 billion rows from 2023 to Sep. 2026.
- Developed a FastMCP server powering conversational analytics in Slack, grounded in dbt docs, Lightdash dashboards, and semantic-layer metadata for self-service data discovery.
- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, eliminating recurring manual effort and human error.
- Built GenBI dashboards and monitoring with Lightdash, dbt, and Airflow for DAG, Athena, and S3; migrated Timo Metrics from Loki/Grafana to SigNoz, consolidating logs and metrics to uphold data SLAs.
- Delivered Tracking247 for the Customer team: Streamlit app with Airflow pipelines syncing shipment status into MongoDB, centralizing updates and cutting manual follow-up.

Projects

Mini Lakehouse — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered deterministic batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt/Lightdash analytics, Modal GPU OCR, and Netdata observability exposed through Cloudflare Zero Trust without public host ingress.

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese web-research pipeline that plans LLM queries, searches across Brave, Tavily, Serper, and Exa, deduplicates and reranks sources, extracts bounded context, and generates source-backed Facebook posts.
- Integrated PydanticAI with an OpenAI-compatible Timo gateway for typed output, validation, and usage tracking; added retrieval budgets, retries, and CLI and Streamlit interfaces.

Technical Skills

Programming	Python, SQL, C++
Data Engineering	Databricks, Spark, PySpark, Airflow, dbt, PyIceberg, Polars, Dagster
Cloud & Infrastructure	AWS (S3, Athena, EMR, Glue), OCI, Cloudflare Zero Trust, Terraform, Docker, Netdata, SigNoz, Git
Lakehouse & Databases	Apache Iceberg, AWS Glue Data Catalog, PostgreSQL, MariaDB, MongoDB
AI & Analytics	Model Context Protocol (MCP), FastMCP, PydanticAI, semantic layer design, Lightdash, Streamlit, Slack

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350